

Response to Reviewers

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Manuscript. A Diagnostic Evaluation of Small Language Models for Offline Writing-Feedback Tutoring in Brazilian Portuguese: A Study on Structural and Pedagogical Adherence under Public-School Infrastructure Constraints

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Decision. A revised version is required for further review, 27 July 2026

Dear Editors, dear Reviewers,

We thank both reviewers for reports that were precise, constructive and, on the central point, convergent. Both pressed on the reliability of the coding that supports the qualitative analysis. Measuring it properly obliged us to withdraw a reliability check we had already reported, rebuild the coding instrument, and run the round again with two coders external to the research team. One claim of the submitted version is now weaker, one is differently framed, and the section the reviewers questioned rests on evidence with defensible provenance. Every other point raised was addressed in the text.

Below, each reviewer comment is quoted in full, followed by what changed and where. Section, table and figure numbers refer to the revised version. A summary table closes the letter.

Reviewer A

A1. The lexical proxy for RQ2

My main concern is the evaluation of the second research question. The use of lexical markers associated with the cohesion taxonomy is a relatively weak proxy for metalinguistic or pedagogical competence. Mentioning an appropriate term does not necessarily mean that the model correctly identifies the problem or helps the student revise it; conversely, a pedagogically useful response may not use the expected terminology. I therefore recommend either strengthening this analysis through independent human evaluation or clearly presenting the keyword analysis as exploratory evidence rather than as a direct measure of pedagogical adherence.

We did both. The keyword analysis is now presented as an exploratory screen, and its construct validity was estimated against independent human judgment.

Two teachers of Portuguese, external to the study (their recruitment and profile are described under A3), judged each of the 39 responses of the conformant model, in a separate pass and against a definition frozen before annotation, on whether the response addresses the student's own text as a linguistic object rather than the content of the literary source. The two coders agree with **each other** at $\kappa = 0.72$ (raw agreement 87.2%). The lexical screen agrees with them at only $\kappa = 0.28$ and 0.43 , and it credits 20 of 39 turns (51.3%) where they credit 26 and 25 (66.7% and 64.1%).

The screen therefore **under-credits** the construct at corpus level, not only turn by turn, which is a stronger limitation than the submitted version declared. The revised manuscript reports the 51.3% figure as a lower bound, always accompanied by the two human figures, in the Abstract, in Section 6 and in the Conclusion, and treats the human judgment as the better-grounded measure of RQ2 (Section 8, first limitation, states this in full). The framing of the finding changes accordingly: not that the model rarely addresses the student's writing, since two experienced readers say it does so in about two thirds of the turns, but that its mediational repertoire is narrow and a substantial share of

its moves rests on material the student's text does not contain (Subsections 7.5 and 7.6).

A2. The role of the human reference corpus

The role of the human reference corpus also needs some clarification. Although the authors state that it is descriptive rather than a gold standard, the comparison between model and teacher frequencies may still be interpreted as a performance benchmark. The manuscript should more clearly define the corpus as an empirical reference for pedagogical mediation practices.

The opening of Section 7 was rewritten to define the corpus in exactly those terms. It now states that the reference is *not* used as a performance yardstick against the model; that the specialist feedback was analyzed to derive the mediation functions that underpin the pedagogical dimension of the benchmark, so that the comparison the study makes is *model against mediation functions derived from expert practice*, not model against teacher; and that no claim of the form "the model is worse than the teachers" is made, because the planted-error design of the scenarios makes problem detection circular by construction. The text also explains why the corpus was kept: its analytical value lies in the mediational layer (when to ask rather than tell, how much scaffolding to extend, whether to model a repair), which the scenario does not dictate.

Table 9 was reframed in the same direction. The teacher column is labelled and captioned as a descriptive reference distribution, explicitly declared non-commensurable with the model columns because it was coded under an earlier version of the instrument, and the table no longer carries a "gap" column that invited reading it as a score. Figure 2 was redrawn so that the reference appears as a grey marker beside the interval spanned by the two coders, rather than as a bar to be matched.

A3. Reliability validation of the coding scheme

In addition, the revised coding scheme would benefit from further reliability validation, particularly for the mediation functions that are central to the conclusions.

This is the point on which the manuscript changed most, and it coincides with Reviewer C's item C9. We describe it once here.

The $\kappa = 0.14$ we first reported is withdrawn. While preparing the validation, we found that the pass it was compared against had been produced programmatically rather than by a human coder, and that the two passes did not apply the same version of the codebook. Such a pair does not estimate inter-rater reliability, so we withdrew the figure instead of letting it stand. Subsection 7.6 states this in the text.

The instrument was rebuilt before the new round. Version 0.5 of the codebook separates the *presence* of a mediation move from its *validity*: a move present in form but resting on a misreading of the student's text is coded as the function present with a false positive attached, not as the function absent. Quoted evidence is mandatory for every function marked and a written justification for every false positive. A dated analysis plan fixed the reporting rules before annotation, including the rule that no mean κ would serve as a decision criterion.

The round that replaces it. Two teachers of Portuguese coded all 39 model turns independently. Neither is an author, a supervisor, or a member of the research group; neither is one of the five teachers who produced the specialist corpus; neither had prior access to the manuscript, the repository, the coding manual, or any earlier coding. Both hold a degree in Letters and a master's in Applied Linguistics, and one currently teaches the 8th and 9th grades the study targets. Both were paid a flat fee agreed in advance, contingent on delivery and not on any result. Subsection 7.6 and the Declarations report the recruitment, the eligibility criteria, the exclusions and the compensation.

Result (Table 10, new), read in three tiers as the analysis plan required:

	functions	reading
Reliable	recognize competence $\kappa = 0.64$; provoke reflection $\kappa = 0.57$; metalinguistic focus $\kappa = 0.72$	carry quantitative description
Not estimable	name the problem, offer a cue, propose revision	1 to 5 occurrences in 39; raw agreement 87–92%, PABAK 0.74–0.85, κ collapses
Absent	partial modeling, reinforce autonomy	0 of 39 under both coders

The middle tier is the kappa paradox of Feinstein and Cicchetti: a corpus with one to five instances of a function cannot support an estimate of its reliability, which is not the same as having shown it unreliable. Every category is reported, with its prevalence under each coder, its raw agreement, its κ and its prevalence-adjusted counterpart.

The disagreement is taxonomic, not perceptual. The two codings differ on 35 decisions across 25 turns, and **31 of the 35 are the same quoted stretch of feedback assigned to a different function**. The criterion is explicit: we test whether the other coder recorded another function in the same turn whose quoted evidence covers the same text, at 80% or more of the shorter quotation's content words. Only four disagreements concern whether a move is present at all. The reassignments concentrate on three boundaries, and reflection against expansion accounts for 14 of the 31: the model returns the same shape every time, a recognition clause followed by two open questions that are simultaneously anchored in the student's text and requests for content it does not contain, and the codebook does not say whether these two functions may be recorded together. Both readings are consistent with the guide, which makes this a defect of the instrument, and one an earlier round had already flagged in the same neighbourhood. **We decline to recompute κ with the categories merged**: the analysis plan forbids adjusting the instrument after seeing the result.

No false-positive rate is published. Agreement on whether a move is unsupported by the student's text is $\kappa = 0.21$ in the largest conditional subset. Asking only whether a turn carries any unsupported move, irrespective of function, gives $\kappa = 0.18$, with one coder flagging 13 of the 39 turns and the other 27: a difference of severity, not of filing. We report only the floor both codings support, namely 11 turns in which both teachers independently recorded a move resting on something the student's text does not contain.

Effect on the manuscript. Subsection 7.5 was rewritten around the two external codings, and Table 9 now shows the prevalence recorded by each coder instead of a single column. Only the reliable tier bears quantitative weight; the argument rests on the shape both codings share: recognition and reflection are recorded at identical rates (92% and 77%), the operational core of the specialist corpus (naming the problem, offering a cue, proposing a revision) appears in at most five turns under either coder, and two functions (modeling a repair, reinforcing autonomy) are empty under both. Section 8 now names the reliability of the model-side coding as the sharpest limitation of the paper and states its new shape.

A4. Scenario-level analysis

The statistical interpretation should also acknowledge more explicitly that the 39 responses per model derive from only 13 scenarios repeated three times. The repetitions are useful for assessing stability, but they do not provide the same diversity as 39 independent scenarios. A scenario-level analysis would strengthen the conclusions.

Section 6 gained a paragraph, "The scenario as the unit of analysis", that repeats the RQ2 analysis with the scenario as the sampling unit. Taking the scenario as the unit, 1 of 13 scenarios adheres in

every repetition (7.7%; Wilson 95% CI [1.4%; 33.3%]) and 12 of 13 in at least one (92.3%; [66.7%; 98.6%]): the model is almost always *capable* of surfacing the metalinguistic object and almost never *reliable* in doing so. Quantifying the clustering directly, the intraclass correlation of adherence is negative ($ICC(1) = -0.11$, one-way ANOVA estimator; design effect 0.79, effective $n = 49.5$), and a cluster bootstrap that resamples scenarios rather than responses ($B = 20,000$) returns [38.5%; 64.1%], marginally narrower than the naive Wilson interval. The repeated-measures design therefore does not inflate the precision of the 51.3% estimate; we nevertheless keep the wider Wilson interval throughout as the conservative reading. The absence of positive intraclass correlation is reported as substantive rather than technical: it is the quantitative form of the instability described in the text, since a model mobilizing the rubric as a stable property of each scenario would necessarily produce positive clustering. The per-scenario breakdown is released with the artifacts.

A5. Causal claims about the origin of the Llama 3.2 behavior

Finally, some causal claims concerning the origin of the Llama 3.2 behavior in instruction tuning or post-training appear stronger than the evidence supports. The experiments demonstrate a robust behavioral pattern under the tested conditions and its reversibility through one-shot prompting, but they do not establish the underlying causal mechanism. The discussion should therefore use more cautious language.

We agree, and the language was changed wherever it appeared. The Abstract no longer says the bias is "inherited from instruction tuning"; it says what the protocol shows, namely that the behavior is invariant to sampling temperature and to the client transport layer, reversible by a single demonstration, and that "the design is behavioural and does not establish the bias's origin". Subsection 5.2 gained a paragraph stating what the evidence does and does not license: the protocol is behavioral and black-box; it establishes that the divergence is systematic (78/78 calls), invariant to temperature (E1) and to transport (E3), shared across two model sizes and reversible by one in-context demonstration (E2, E2b); it does not observe training data, reward models or post-training procedures and therefore cannot attribute the behavior to instruction tuning, preference optimization or any other stage of the pipeline. The alignment with the format-bias literature is offered as the most parsimonious reading of a behavioral regularity, not as a demonstrated mechanism, and the text says what would be needed to establish one (access to the training pipeline or controlled ablations over checkpoints).

A6. Restriction of the insufficiency conclusion to the evaluated setting

Similarly, the conclusions about the insufficiency of off-the-shelf SLMs should be explicitly restricted to the evaluated zero-shot, local inference setting.

The restriction is now explicit at every point where the conclusion is stated. The Abstract reads: "in the regime evaluated here (zero-shot prompting, local CPU-only inference, Q4 quantization, no grammatical coercion), off-the-shelf generalist models are insufficient", and closes with "the finding characterizes an operating regime rather than a fixed limit", since a single in-context demonstration reverses the structural failure. Subsection 5.3 states that the structural finding is a property of the zero-shot, no-coercion deployment regime, and Section 8 carries a "Generalization boundary" limitation binding all results to the evaluated families, their Q4 quantization, the Ollama runtime and the zero-shot regime. The Conclusion restates the boundary when returning to each research question.

Reviewer C

C1. Organization paragraph of the Introduction

Inconsistency in the text's organization in the Introduction: the final paragraph of Section 1 describes the article's organization and jumps from Section 7 directly to Section 9 ("Section 7 introduces a specialist human reference (...) and Section 9 synthesizes conclusions"), omitting any explicit mention of Section 8 ("Threats to Validity"), which does exist and is substantial.

Corrected. The paragraph now reads "...Section 7 introduces a specialist human reference and describes the repertoire of mediation functions that grounds the pedagogical construct; Section 8 states the threats to validity that bound the scope of the claims; and Section 9 synthesizes conclusions...".

C2. Call count of the falsification protocol and the E3 isolation

Completeness gap in Section 3.8/5.3: the reported total of 80 calls in the falsification protocol does not reconcile with the explicit breakdown into E1 (26), E2 (24), and E2b (24), which sum to 74; the exact number of E3 calls (the curl-based isolation) is never stated, nor does it appear as a k/n count in Table 4 (which reports only "failure reproduced" qualitatively for E3). I suggest revising this count and specifying the number of E3 calls, should the authors choose to keep the reported totals.

The reviewer is right, and the count was wrong rather than incomplete. The E3 isolation was re-executed in full and persisted as a released artifact: 26 calls (2 models × 13 scenarios × 1 repetition) issued through curl directly to the Ollama endpoint, with no mediation from the Python client. The correct total is therefore **100 calls (E1 26, E2 24, E2b 24, E3 26)**, and the breakdown is now written out in Subsection 3.8. Table 4 reports E3 as a k/n count on the same footing as the other rows, 0/13 and 0/13, with its Wilson interval and Fisher test, instead of the qualitative "failure reproduced". The figure of 80 also appeared in two places the reviewer did not mention, the Abstract and the Data Availability statement; both now read 100.

C3. Textual density and repetition of figures

High textual density: some paragraphs are quite long, with multiple subordinate clauses, and statistics are repeated almost verbatim in several places (the Abstract, Section 6, Section 7.5, and the Conclusion all repeat the same figures of 51.3% and 3%/15%/0% vs. 80%/69%/57%). This does not create confusion, but it demands more reading effort than desirable for an already long article.

The repeated trio was reduced. The pairing of model rates against the reference prevalences of 80%/69%/57% now appears once in prose, in Subsection 7.5, and in Table 9; it was removed from the Abstract and from the Conclusion. The 51.3% figure is kept only where the argument needs it, and always beside the two human figures that now qualify it (see A1).

We should be candid about the net effect on length. The reliability round both reviewers asked for added a new subsection (7.6), a new table (Table 10) and a longer limitations section, so the revised manuscript is longer than the submitted one despite the cuts. We took the view that reporting the reliability evidence in full, including its unfavourable parts, was the reviewers' priority, and we would welcome editorial guidance if the journal prefers part of Subsection 7.6 moved to an appendix.

C4. Table 5 layout

Table 5 appears to extend beyond the page's right margin; please check the layout.

Fixed. Table 5 now uses two fixed-width paragraph columns sized to the column width, with a smaller font, and no longer overflows the margin.

C5. Brazilian literature on automated essay evaluation

I suggest expanding Section 2 with at least one reference to the Brazilian literature on automated writing evaluation/essay correction, for example, the body of research on automated scoring of ENEM essays, a natural and expected comparison for an article defined by its focus on "Brazilian Portuguese writing" and "basic education", or explicitly justifying this absence.

Section 2 gained a fourth front on the Brazilian tradition of automatic essay evaluation. It cites Essay-BR (Marinho et al., 2022), the public corpus of ENEM argumentative essays scored on the five competencies, and Barbosa and Mauá (2026), who report cross-prompt, trait-specific scoring over 38 prompts with QWK between 0.60 and 0.73. The contrast with the present study is written out rather than left implicit: that tradition addresses *summative* evaluation (a grade for a finished text, over authentic corpora, typically with server-class compute), whereas this study addresses *formative* return (no grade, a structured intervention meant to lead the student back into the text, offline on school hardware). The two are different tasks, and the text explains why agreement coefficients on score prediction have no counterpart in what we measure.

C6. A writing or language-tutoring benchmark for context

The two pedagogical benchmarks cited as theoretical support (MathTutorBench, Macina et al., 2025; GuideEval, Liu et al., 2025) belong to adjacent domains (mathematics; general Socratic tutoring), but no writing/language-tutoring benchmark in another language is discussed to contextualize whether the 51.3% rate (Table 5) or the gaps in Table 9 are high, low, or comparable to findings published in similar settings.

Section 2 now discusses TutorBench (Srinivasa et al., 2025), a rubric-graded tutoring benchmark in which no frontier model exceeds 56% overall and the "assessment and feedback" use case averages roughly 51% across models. The proximity to our rate is noted and immediately bounded: TutorBench scores tutoring quality in English with frontier-scale models and a judge model, whereas our figure is a lexical screen over a 3-billion-parameter model running locally in Brazilian Portuguese, so the comparison is one of order of magnitude and direction only. Its value is to show that partial performance on the mediational dimension is the prevailing finding in the literature rather than an artifact of our small models or of our language.

C7. Prior work on local SLM execution in Brazilian or Latin American education

There is no discussion of prior work already using Ollama/local SLM execution specifically in a Brazilian or Latin American educational context, which would help position the novelty of the "public-school infrastructure" framing relative to related efforts, if any exist.

We searched for such work and found the closest match to be Chan et al. (2026), who evaluate open Llama models from 1B to 70B for locally hosted school settings and report the throughput and concurrency penalties that grow with model size, arriving from an infrastructure angle at a conclusion convergent with ours. Section 2 now cites it and marks the differences (multi-GPU configurations, general assistant use). For the Brazilian and Latin American context specifically, the revised text states the gap explicitly: we are unaware of prior work evaluating locally executed SLMs for a pedagogically specified task in Brazilian Portuguese under CPU-only school hardware, and that is the gap this benchmark addresses. We would be grateful for any pointer the reviewer may have.

C8. Prominence of the RQ2 caveat

The Abstract and the Conclusion already qualify the RQ2 finding as a "corroborating qualitative probe rather than a co-equal claim" and "not as an independent claim of equal evidentiary weight," which is a positive practice. Even so, this qualification appears diluted within subordinate clauses, while the figures (51.3%; 3%/15%/0% vs. 80%/69%/57%) are presented assertively elsewhere in those same passages. I suggest giving this caveat more prominence, so that a reader who reads only the Abstract does not retain the figures without their proper

qualification.

The caveat was promoted to a sentence of its own, placed *before* the figures. The Abstract now reads: "We state its evidentiary status first: this is an exploratory single-model analysis whose instrument is a lexical screen over Koch's terminology, so its figures corroborate the diagnosis rather than carry it." Only then does the 51.3% appear, and it is followed in the same sentence by the two human figures and the agreement coefficients that qualify it (A1). The mediation-function result is likewise introduced in the Abstract as "a pattern and not a measurement", and the 3%/15%/0% trio was removed from the Abstract and the Conclusion (C3).

C9. A second blind coding pass of the model's 39 outputs

The authors already report, as an explicit limitation (Sections 7.5 and 8), the absence of a second blind coding pass of the model's 39 outputs against the eight mediation functions (Table 9). Given that these figures are the most frequently cited in the article, I suggest the authors consider actually carrying out this second coding pass and reporting the corresponding κ , rather than merely flagging it as future work.

It was carried out, with two independent external coders rather than one, over the full corpus of 39 turns, under a rebuilt instrument and a pre-registered analysis plan. The design, the result per function, the divergence analysis and the effect on Table 9 are described under A3 above and in Subsection 7.6 and Table 10 of the manuscript.

C10. DOIs in the reference list

Please review the reference list: at least eight entries lack a DOI, although they typically have one available via arXiv (for example, Abidin et al., 2024; Dong et al., 2024; Gemma Team, 2024; Grattafiori et al., 2024; Liu et al., 2025; Willard and Louf, 2023; Wu and Aji, 2023; and Yang et al., 2024).

Fixed, with a note on the cause. All eight entries already carried a doi field in the bibliography source; the journal's bibliography style prints the DOI for @article and @inproceedings entries but not for the @misc type under which arXiv preprints are recorded, so the field was silently dropped at typesetting. We added the DOI explicitly to each of the eight entries so that it renders in the reference list. Of the four references added in this revision (C5 to C7), three carry a DOI; the PROPOR 2026 paper (Barbosa and Mauá) has no DOI assigned in the ACL Anthology, so its persistent Anthology URL is given instead.

Changes not requested by the reviewers

Three further changes follow from the reliability round and are reported for completeness.

Annotator independence. The submitted version described the reliability pass of the specialist corpus (Subsection 7.4, $\kappa = 0.83$ over 20 turns) as performed by "a second, blind, and independent annotator". That was too generous: the second coder was blind to the first coding but is one of the authors. The text now says so and the CRediT statement records it. That figure is not comparable to the model-turn round of Subsection 7.6 (different codebook versions, different corpora, different coders), and the text no longer invites the comparison.

Ethics, compensation and released material. The Declarations now state that the two external coders acted as paid specialist collaborators over fictional material, not as research subjects; that no personal or sensitive data about them is published; that each signed a service agreement covering the fee and the use of the coding in this paper, in the doctoral thesis it belongs to, and in the open release of the instrument and coded data; and that they are identified only as coder A and coder B. The funding statement records that the fees were paid personally by the first author. The

Data Availability statement lists the released material: the version 0.5 protocol and coder's guide, the annotation package delivered to each coder, both completed codings with quoted evidence and written justifications, the provenance record of the captured spreadsheets, the completeness audit, the analysis scripts, the item-by-item divergence listing, and the dated analysis plan.

Figure 2 was redrawn to show, for each mediation function, the interval spanned by the two coders rather than a single bar, with the specialist reference as a separate marker.

Summary of changes

Item	Reviewer	Where in the revised manuscript
Independent human evaluation of RQ2; screen reported as lower bound	A1, C8	Abstract; §6; §7.6; §8; §9
Corpus defined as empirical reference, not yardstick	A2	§7 opening; Table 9 caption; Figure 2
Reliability of mediation-function coding; $\kappa = 0.14$ withdrawn; new external double coding	A3, C9	§7.5; §7.6 (new); Tables 9 and 10; §8
Scenario-level analysis (ICC, design effect, cluster bootstrap)	A4	§6, paragraph "The scenario as the unit of analysis"
Causal language softened; black-box scope stated	A5	Abstract; §5.2
Insufficiency conclusion restricted to the evaluated regime	A6	Abstract; §5.3; §8; §9
Roadmap paragraph mentions Section 8	C1	§1
Falsification count corrected to 100; E3 = 26 calls, k/n in Table 4	C2	Abstract; §3.8; Table 4; Data Availability
Repeated figures reduced	C3	Abstract; §7.5; §9
Table 5 layout	C4	Table 5
Brazilian AES literature (Essay-BR; Barbosa and Mauá)	C5	§2
Writing-tutoring benchmark for context (TutorBench)	C6	§2
Local SLM execution in school settings (Chan et al.); gap stated	C7	§2
DOIs rendered for the eight arXiv entries	C10	References
Annotator independence corrected; CRediT updated	(authors)	§7.4; Declarations

Item	Reviewer	Where in the revised manuscript
Ethics, compensation, released material	(authors)	Declarations; Data Availability

We are grateful to the reviewers for pressing on the reliability point. The manuscript is materially stronger for it, and more transparent about where its evidence is thin.

Sincerely,

Randerson Oliveira Melville Rebouças, on behalf of the co-authors